

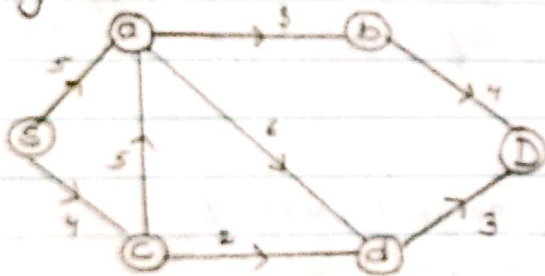
Network Flow Problems Flow Network

A flow network or transport network is a connected directed graph $G=(V,E)$ that does not contain any loops such that,

- There are exactly two distinguished vertices S and D , called source and destination (sink) of graph (G) respectively.
- There is a non-negative real values function k defined on edge E called the capacity function of G .

Then, (G,k) is called transport network and function k is called capacity function of G . The vertices distinct from S and D are called intermediate vertices. If $e \in E$ then $k(e)$ is called capacity of e .

For eg :

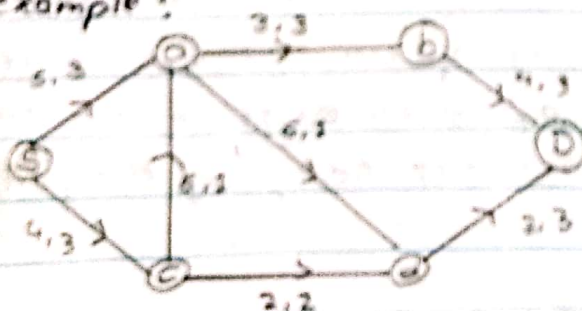


Flow: let (G,k) be a transport network, the a flow in G is a non-negative function f defined on edge E such that,

$$0 \leq F(e) \leq k(e) \text{ for each edge } e \in E$$

where $F(e)$ = flow to the edge e .

For example :



DAG
→ Directed Acyclic Graph

Network Flow Problems

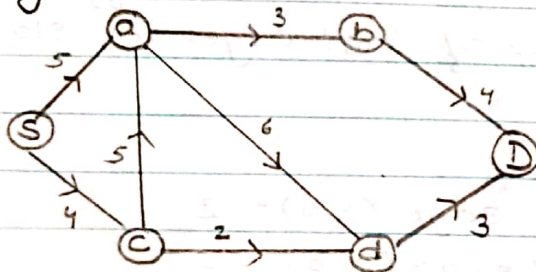
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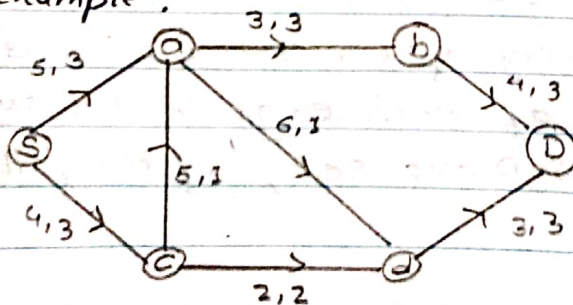


Flow: let (G, k) be a transport network, then a flow in G is a non-negative function f defined on edge E such that,

$$0 \leq f(e) \leq k(e) \text{ for each edge } e \in E$$

where $f(e) = \text{flow to the edge } e$.

For example :



From the above figure, the flow-in of vertex 'a' is
 $F(s, a) + F(c, a) = 3 + 1 = 4$

The flow-out of vertex 'a' is
 $F(a, b) + F(a, d) = 3 + 1 = 4$ and so on.

Saturated and unsaturated edges

The edges for which the flow and capacity are equal called saturated edges. Otherwise, they are called unsaturated edges.

From the above graph,

edges (a, b), (c, d) and (d, D) are saturated and remaining others are called unsaturated edges.

Slack of edge:

The slack of edge 'e' is the difference of capacity and maximum flow of that edge. The slack of saturated edge is 0.

For example,

$$\text{The slack of edge } (s, a) = 2$$

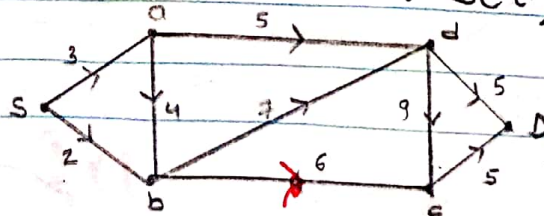
$$\text{Slack of edge } (a, b) = 0$$

S-D cut:

In a flow network, an S-D cut is a cut, that requires the source 'S' and sink 'D' to be in different subsets and it consists of edges going from the source side to sink side (destination).

The capacity of an S-D cut is defined by the sum of capacity of each edge in the cut set.

For eg: Find the S-D cut set, of the following graph



Soln,

x	x'	Capacity(x, x')
$\{s\}$	$\{a, b, c, d, 0\}$	$2 + 3 = 5$
$\{s, a\}$	$\{b, c, d, 0\}$	$5 + 4 + 2 = 11$
$\{s, b\}$	$\{a, c, d, 0\}$	$3 + 7 = 10$
$\{s, c\}$	$\{a, b, d, 0\}$	$6 + 3 + 2 + 5 = 16$
$\{s, d\}$	$\{a, b, c, 0\}$	$3 + 2 + 5 + 9 = 19$
$\{s, a, b\}$	$\{c, d, 0\}$	$5 + 7 = 12$
$\{s, a, c\}$	$\{b, d, 0\}$	$5 + 5 + 5 + 2 + 4 = 21$
$\{s, a, d\}$	$\{b, c, 0\}$	$9 + 5 + 4 + 2 = 20$
$\{s, b, c\}$	$\{a, d, 0\}$	$3 + 7 + 5 + 4 = 19$
$\{s, b, d\}$	$\{a, c, 0\}$	$3 + 7 + 5 + 4 = 19$
$\{s, a, b, c\}$		
$\{s, a, b, c, d\}$		
$\{s, a, b, d\}$		
$\{s, b, c, d\}$		
$\{s, a, c, d\}$		

Flow augmenting path

A flow-augmenting path from source s to sink D is a sequence of edges from s to D such that, for each edge in this path, the flow $F(e) < k(e)$ on forward edges and $F(e) > 0$ on backward edges.

It means that such a path is not optimally used and it can transfer more units than it is currently transferring. If the flow for at least one edge of path reaches its capacity, then obviously the flow can be augmented.

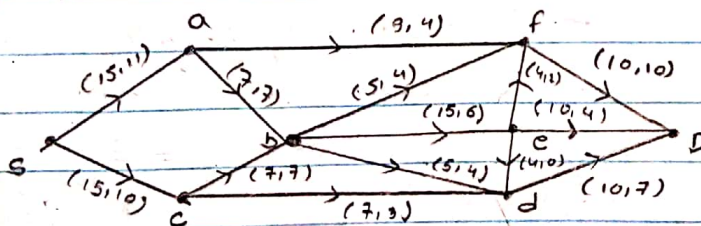
Max-Flow min cut:

In any network, the value of the max flow is equal to the value of min cut.

This theorem asserts the following:

- i) The existence of a minimal cut (x, x')
- ii) The existence of maximal flow F .
- iii) The equality of F and $c(x, x')$ for any maximal flow F and any minimal cut (x, x')

Q. Find the maximum flow for the following Network graph.



Soln,

At first we see for edges that are unsaturated and has forward paths only.

Here, unsaturated forward path

$$P_1 = s - c - d - D$$

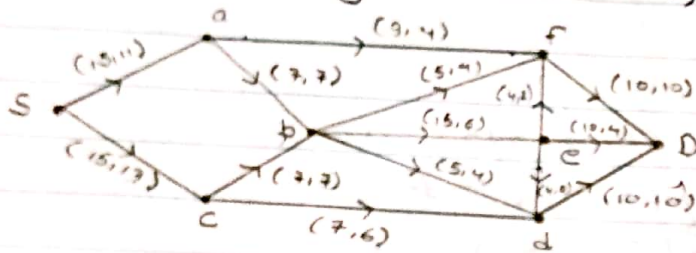
Now,

$$\text{Slack of } (s, c) = 15 - 10 = 5$$

$$\text{Slack of } (c, d) = 7 - 3 = 4$$

$$\text{Slack of } (d, D) = 10 - 7 = 3$$

Minimum slack value is 3, so, add 3 to each edges, so, above graph becomes,



Now, look for edges which are unsaturated and has backward edges.

$$P_2 = S-a-f-e-D$$

$$P_3 = S-c-d-e-D$$

$$P_4 = S-c-d-b-e-D$$

$$P_5 = S-a-f-b-e-D$$

For P_5 : Slack of $(s, a) = 15 - 11 = 4$

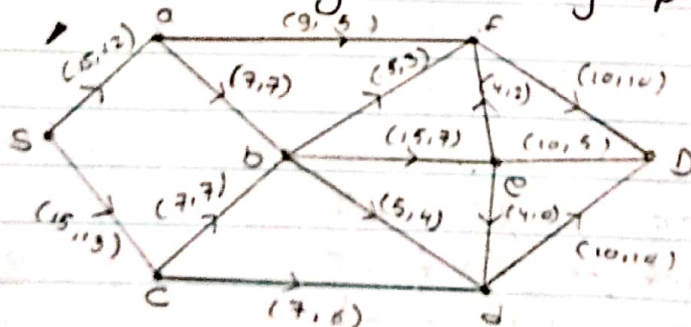
$$\text{Slack of } (a, f) = 9 - 4 = 5$$

$$\text{Slack of } (f, b) = 5 - 4 = 1$$

$$\text{Slack of } (b, e) = 15 - 6 = 9$$

$$\text{Slack of } (e, D) = 10 - 4 = 6$$

Now, add 1 to forward edges and subtract to backward edges. The graph becomes,



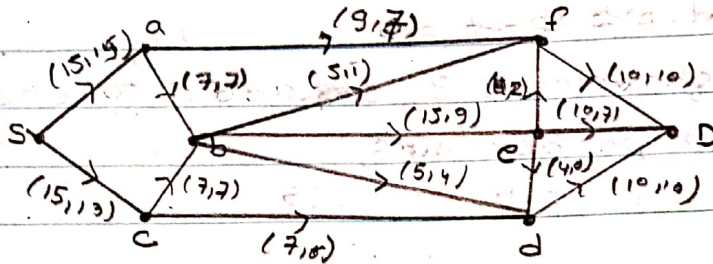
Again, Slack of $(s, a) = 3$

$$\text{Slack of } (a, f) = 4$$

$$\text{Slack of } (f, b) = 2$$

Slack of $(b,e) = 8$

Slack of $(e,d) = 5$



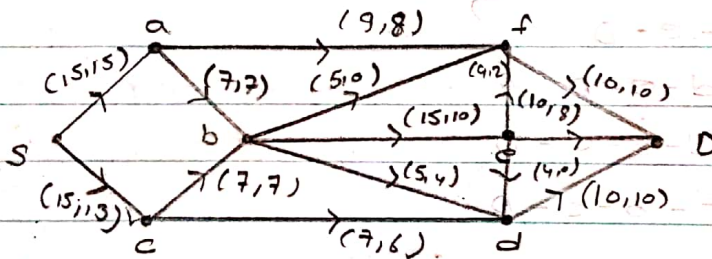
Again, Slack of $(S,a) = 15 - 14 = 1$

Slack of $(a,f) = 9 - 7 = 2$

Slack of $(f,b) = 5 - 1 = 4$

Slack of $(b,e) = 15 - 9 = 6$

Slack of $(e,d) = 10 - 7 = 3$



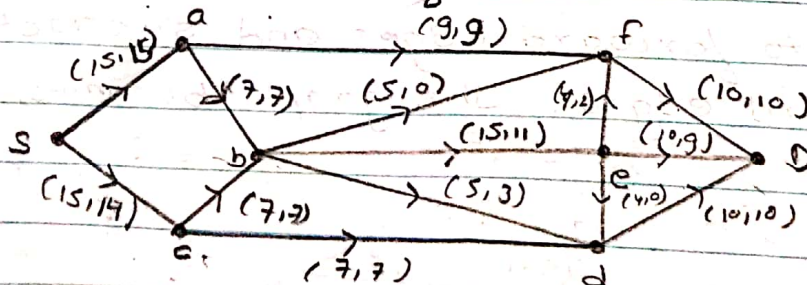
For P_4 : Slack of $(S,c) = 15 - 13 = 2$

Slack of $(c,d) = 7 - 6 = 1$

Slack of $(d,b) = 5 - 4 = 1$

Slack of $(b,e) = 15 - 10 = 5$

Slack of $(e,d) = 10 - 8 = 2$



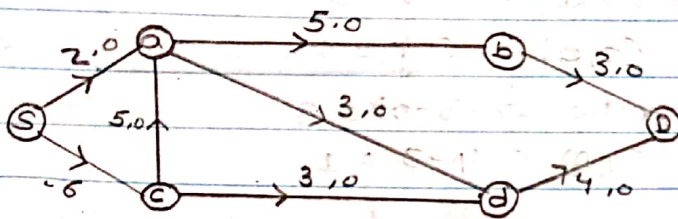
For P_2 : Slack of $(S,a) = 4$

Slack of $(a,f) = 3$

Slack of $(f,e) = 4$

Slack of $(e,d) = 10$

Q. Find max flow of the following network flow graph



Soln:

At first we see for edges that are unsaturated and has forward paths only.

Here, unsaturated forward path

$$P_1 = S-a-b-D$$

$$P_2 = S-c-d-D$$

$$P_3 = S-c-a-d-D$$

$$P_4 = S-a-d-D$$

$$P_5 = S-c-a-b-D$$

Now, for P_1 :

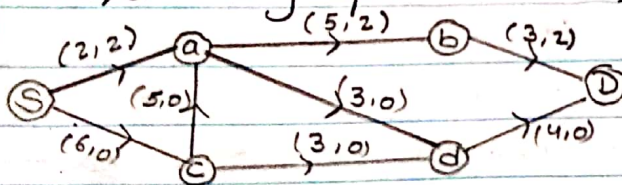
$$\text{Slack of } (S, a) = 2 - 0 = 2$$

$$\text{Slack of } (a, b) = 5 - 0 = 5$$

$$\text{Slack of } (b, D) = 3 - 0 = 3$$

Minimum slack value is 2. So add 2 to each edges.

So, above graph becomes,

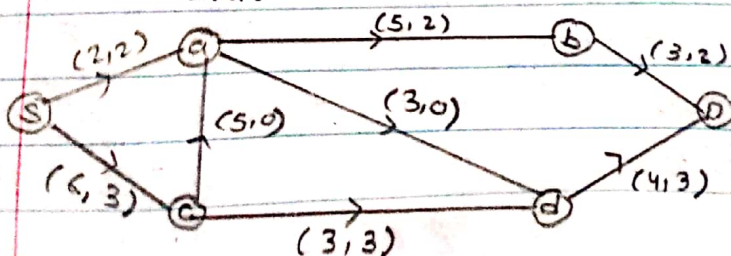


For P_2 :

$$\text{Slack of } (S, c) = 6 - 0 = 6$$

$$\text{Slack of } (c, d) = 3 - 0 = 3$$

$$\text{Slack of } (d, D) = 4 - 0 = 4$$



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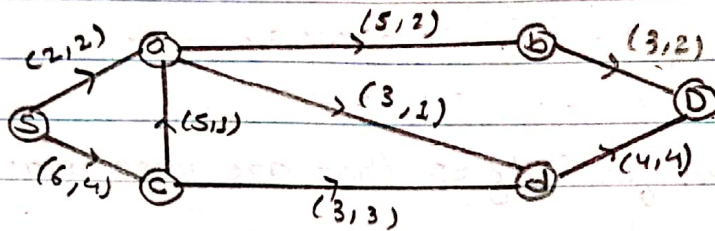
For P3:

$$\text{Slack of } (s, c) = 6 - 3 = 3$$

$$\text{Slack of } (c, a) = 5 - 0 = 5$$

$$\text{Slack of } (a, d) = 3 - 0 = 3$$

$$\text{Slack of } (d, D) = 4 - 3 = 1$$



For P5:

$$\text{Slack of } (s, c) = 6 - 4 = 2$$

$$\text{Slack of } (c, a) = 5 - 1 = 4$$

$$\text{Slack of } (a, b) = 5 - 2 = 3$$

$$\text{Slack of } (b, d) = 3 - 2 = 1$$

